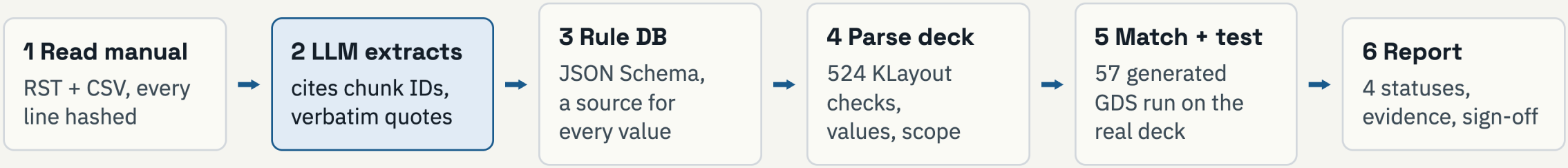
PROTOTYPE · OPEN-SOURCE GF180MCU PDK

AI-assisted PDK rule checking for GF180MCU

An LLM reads the design manual. Deterministic code checks the LLM, matches every rule against the shipped KLayout DRC deck, and proves each gap with generated test layouts.



Results on 20 rules

13	4	2	1
MATCH	REVIEW	PARTIAL	MISSING

46 of 57 test layouts behave as the manual requires; all 11 disagreements are explained. **160/160** provenance and grounding checks pass. GitHub Actions reruns it all and gets the same result.

Potential gaps for expert review

Rule	Status	Finding
MT.2b	MISSING	No deck check found for wide-MetalTop spacing
MT.1, MT.2a	PARTIAL	Nothing checked on the 11K MetalTop option
M1.2b	REVIEW	Two wide Metal1 plates too close: never flagged
M1.4	REVIEW	Manual needs over 30%; deck fails only under 30%
PL.5b	REVIEW	Same code as PL.5a: “related” COMP is lost

Automated findings; some may be intentional or already known. Confirmation by PDK experts needed.

github.com/fakhter-uvm014/ai-pdk-gf180mcu · manual @de3240d · deck @9f992d5 · KLayout DRC